

CW Keyers

Anders Helmersson
SM5KAE
email: anders@helmersson.one

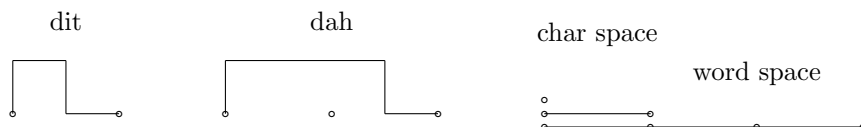
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Abstract

The *Dactylic* keyer is based on the observation that dits are more common than dahs. In order to reduce the movements of the paddle, dits are produced until the paddle is released. If the paddle is alternated from one side to the other, a dah is inserted. A dah is also produced on the first press on the dah side of the paddle.

1 The Morse code

The Morse code is built up from two primitive elements: dit and dah. The dah is twice as long as the dit. A sequence of dits and dahs can be followed by a character space (one dit long) or a word space (three dits long).



2 Paddles and keyers

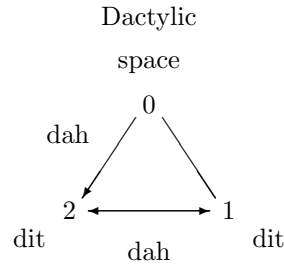
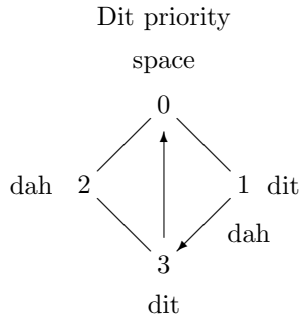
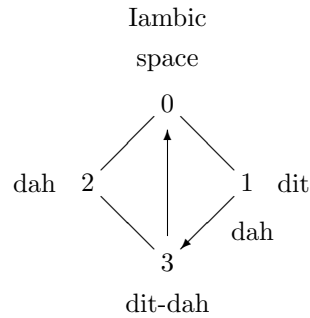
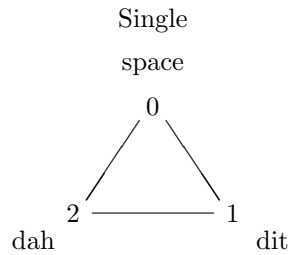
When sending CW code, we often use a key or manipulator, with one or two paddles. A single paddle has three positions: space (0), dit (1) and dah (2). The critical timing applies to the short spaces and dits, the timing for dahs is more relaxed. The *Iambic* key has two paddles: one for dits and the other for dahs. If both paddles are pressed, or squeezed, it becomes position 3, which produces the iambic sequence dit-dah.

In the diagrams below, different keyers are illustrated. The *Single* keyer produces spaces, dits and dahs only according to the position of the paddle. All other keyers add some element when the paddle is *moved*.

The Iambic keyer produces dit-dahs when both keys are pressed (squeezed); a single dah is inserted when moving from 1 to 3

The *Dit-priority* keyer produces dits when both keys are pressed; a single dah is inserted when moving from 1 to 3. The *Ultimatic* and *Dah-priority* keyers are similar, but have a different behavior in the squeezed position (3).

The Dactylic keyer (<https://github.com/ahelmersson/dactylic>) is different. Since dits are more common than dahs, dits are produced when the paddle is kept in its current position until its release. It produces a dah when alternating the paddle from one side to the other. A dah is also sent at the first press on the dah side of the paddle.



3 Comparing keyers

In order to evaluate the keyers we need some statistical model of the Morse code that we send. Two models are used here.

3.1 Maximizing information

Let us first look at how to maximize the information that can be transferred in a Morse message. We achieve this by choosing the probabilities for each element according to its length:

$$\begin{array}{llll} \text{dit:} & p & \text{dit-space:} & p^2 \\ \text{dah:} & p^2 & \text{dah-space:} & p^3 \end{array} \quad \begin{array}{ll} \text{dit-space-space-space:} & p^4 \\ \text{dah-space-space-space:} & p^5 \end{array}$$

All these should sum up to 1. Thus, $p + 2p^2 + p^3 + p^4 + p^5 = (p + p^2)(1 + p + p^3) = 1$, which results in $p = 0.44859$. We also conclude that dits are more than twice ($1/p$) as common as dahs, and the probability for spaces is somewhere in between. The maximum information transferred is 1.1565 bits per time unit.

3.2 Evaluation

When comparing keyers, often only the number of paddle presses counts. Here I use paddle *movements*.

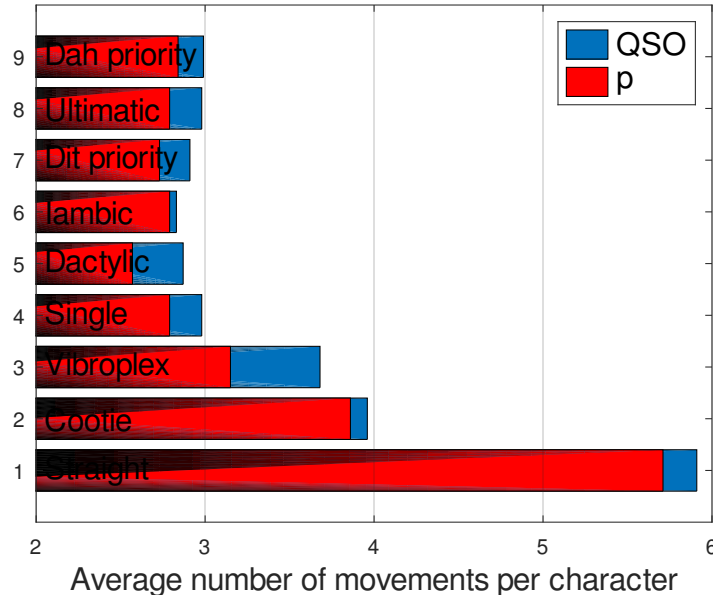
Using an Iambic keyer, you need two presses, but four movements to produce the letter D: press dah, press dit, release dah, release dit. All these four movements counts and they must be accurately timed. If you are late with the first press of dah, a word space is inserted, if you are late with the press of dit, the letter Z is produced, if you are late with the release of dah a C is sent and if you are late with the release of dit you get a B.

Below, I evaluate the keyers using movements. I count one movement for pressing the paddle and one movement for its release. Alternating a single paddle from one side to the other, also counts as one movement. The table shows the average number of movements per character or symbol. The second and third columns give the average number of movements with the element probabilities that maximizes the transferred information. The fourth column is based on statistics using QSO messages from the Morse code trainer and QSO generator by Joe Dellinger, Eric S. Raymond, Thomas Horsten and others.

Keyer/element	Probability/average	$p = 0.45$	QSO
Dit	p	0.45	0.44
Dah	p^2	0.20	0.31
Space	$1 - p - p^2$	0.35	0.25
Symbol length	$2 + \frac{2p+2p^2+p^3}{(1-p-p^2)(1+p)}$	4.74	5.19
Number of elements	$2 + \frac{p+p^2}{1-p-p^2}$	3.86	3.97
Straight	$2 + \frac{2p+2p^2}{1-p-p^2}$	5.71	5.93
Cootie	$2 + \frac{p+p^2}{1-p-p^2}$	3.86	3.97
Vibroplex bug	$2 + \frac{2p^2}{1-p-p^2}$	3.15	3.68
Single	$m(p) := 2 + \frac{2p^2}{(1-p-p^2)(1+p)}$	2.79	2.99
Dactylic	$2 + \frac{p^2}{1-p-p^2}$	2.57	2.86
Iambic	$m(p)$	2.79	2.83
Dit priority	$m(p) - \frac{p^3}{1+p}$	2.73	2.90
Ultimatic	$m(p)$	2.79	2.99
Dah priority	$m(p) + \frac{p^3}{(1+p)^2}$	2.84	2.99

The symbol length and number of elements include the trailing character space.

Keyer comparison



3.3 The letter O and others

One reason for the difference in number of movements for the two statistical models (last two columns in the table), is the high frequency of symbols including consecutive dahs. The most prominent is the letter O (Oscar). The code (dah-dah-dah-space) is seven units long, including the trailing space. The letter O is frequent in English and should have had a shorter code. In addition, common digits like zero (0), one (1) and nine (9), also have long codes (11 and 10 units).

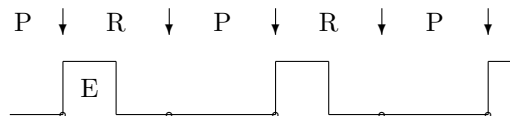
Historically, the Morse code was probably based on reducing the number of elements rather than the length of the code.

4 Dactylic keyer timing

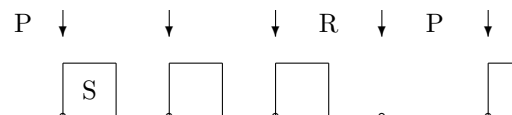
The timing logic tries to spread out the decision intervals as evenly as possible. In the pictures below the decisions are taken at the arrows depending on the position of the key/paddle.

During each interval the keyer looks for changes in the position of the key/paddle relative to the element that is being sent. This is performed using a simple state machine, which is sampled at decision point.

The letter E requires the fastest movements. For sending a sequence of Es, we need to make a movement every time unit: a press, P, during the first interval and a release, R, in the second. This applies to all common keyers.



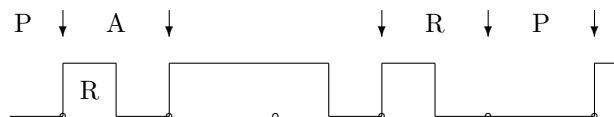
For the letter S, the decisions are taken as late as possible for each element. We press the dit paddle to start sending. Then, we hold the dit paddle until the beginning of the last dit and release. If we release too early, we send the letter I, and if we release too late, we get an H. We must be accurate within one time unit.



A correctly timed character space is inserted automatically by the keyer, but the word space typically needs manual timing.

4.1 Late decisions

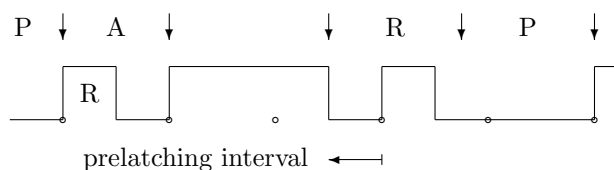
If the decisions are taken as late as possible, just before the next element begins, we get the following timing for the letter R. The interval during sending the dah is twice as long as the remaining intervals. Here A denotes alternating the paddle from one side to the other.



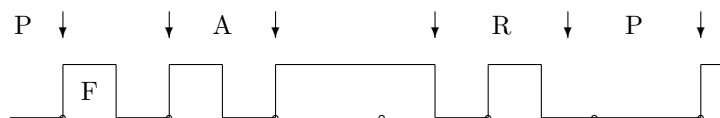
Here we need to release the paddle within an interval of one time unit and again press the paddle within an interval of one time unit.

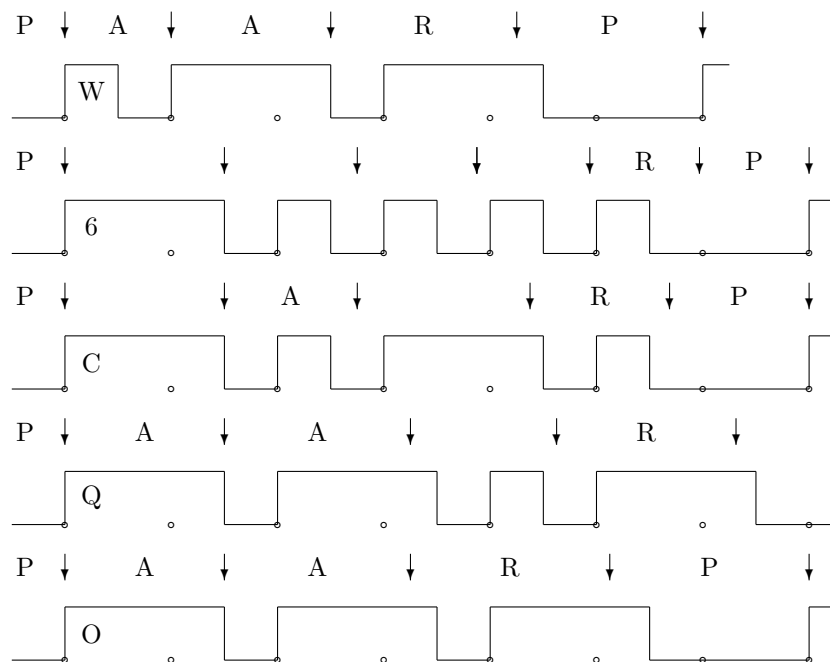
4.2 Relaxed timing

We can spread out the intervals more evenly. For the letter R, we move the decision half a time unit earlier compared to late decisions. The interval during the dah is shortened, while the interval for the release is increased.



The general logic is as follows. The paddle is latched at a prelatching interval before the beginning of the next element. This prelatching interval is reset at the beginning of each symbol, when a space is detected. Otherwise, it is half the prelatching interval of the previous element; if the present element is a dah, we increase the interval by half a time unit. (The prelatching interval is always less than one time unit.) Some examples follow.

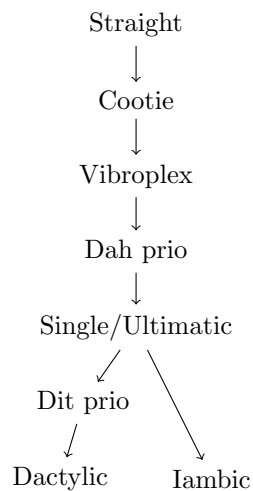




5 Conclusions

The differences between the keyers are quite small. For most keyer, you need less than three movements per character on average. In cootie mode you need four movements and with a straight key, slightly less than six movements.

We can grade the keyers relatively as below



The relative ranking of Iambic and the Dactylic and Dit prio depends on the statistics of the code you send. Choose keyer according to your personal preferences! If you are used to one keyer, there is no reason for changing it to another.